

BI Practical Lab Assignment

Bangladesh Remittances Data Analysis and Visualization using Microsoft Power BI

Aim

To import, clean, transform, model, and visualize the **Bangladesh Remittances Dataset (1995–2025)** using Microsoft Power BI and create an interactive dashboard for analyzing remittance trends.

Objective

1. To obtain a real-world Bangladesh remittance dataset from Kaggle.
2. To import the Excel dataset into Power BI Desktop.
3. To clean and transform the dataset using Power Query.
4. To remove blank and duplicate records.
5. To create separate tables for different analytical requirements.
6. To establish relationships between the tables using common fields.
7. To analyze yearly remittance amounts and growth.
8. To create suitable charts, cards, and interactive visuals.
9. To develop a meaningful Power BI dashboard.
10. To understand the complete Business Intelligence workflow from raw data to visualization.

Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM recommended
- Internet connection

Software

- Microsoft Power BI Desktop
- Microsoft Excel
- Web browser such as Chrome/Edge
- Kaggle account or access to the dataset

Dataset

Bangladesh Remittances Dataset (1995–2025)

The dataset contains annual remittance inflows to Bangladesh, including:

- Year
- Remittances (million USD)
- Remittances (billion BDT)
- YoY Change (%)
- Cumulative Growth vs. 1995–1996 (%)

Theory

Business Intelligence (BI) is the process of collecting, transforming, analyzing, and visualizing data to generate useful information for decision-making.

Microsoft Power BI is a Business Intelligence and data visualization platform that allows users to connect to different data sources, clean and transform data, create relationships between tables, perform calculations, and build interactive reports.

The general BI workflow followed in this practical is:

Data Source → Data Import → Data Cleaning → Data Transformation → Data Modeling → Visualization → Dashboard → Insights

Power Query

Power Query is used for extracting, cleaning, and transforming data before it is loaded into the Power BI data model.

Data Modeling

Data modeling involves organizing tables and defining relationships between them using common fields such as **Year**.

Data Visualization

Power BI provides different visualizations such as:

- Cards
- Column/Bar charts
- Funnel charts
- Pie/Donut charts
- Tables
- Slicers

These visuals help identify trends, comparisons, growth patterns, and important insights.

Procedure / Steps

Step 1 – Find and Download the Dataset

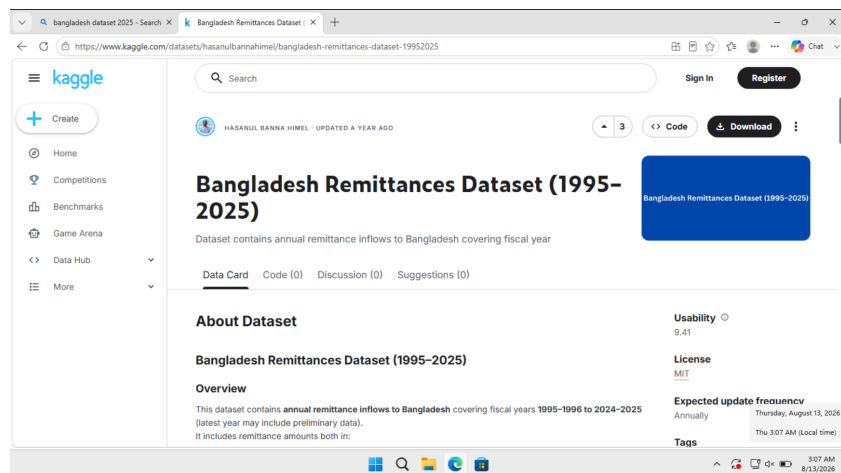
The **Bangladesh Remittances Dataset (1995–2025)** was selected from Kaggle.

The dataset contains annual remittance inflows to Bangladesh from fiscal year **1995–1996 to 2024–2025**.

The dataset was downloaded in **XLSX format**.

Output:

Bangladesh Remittances Dataset (19952025).xlsx

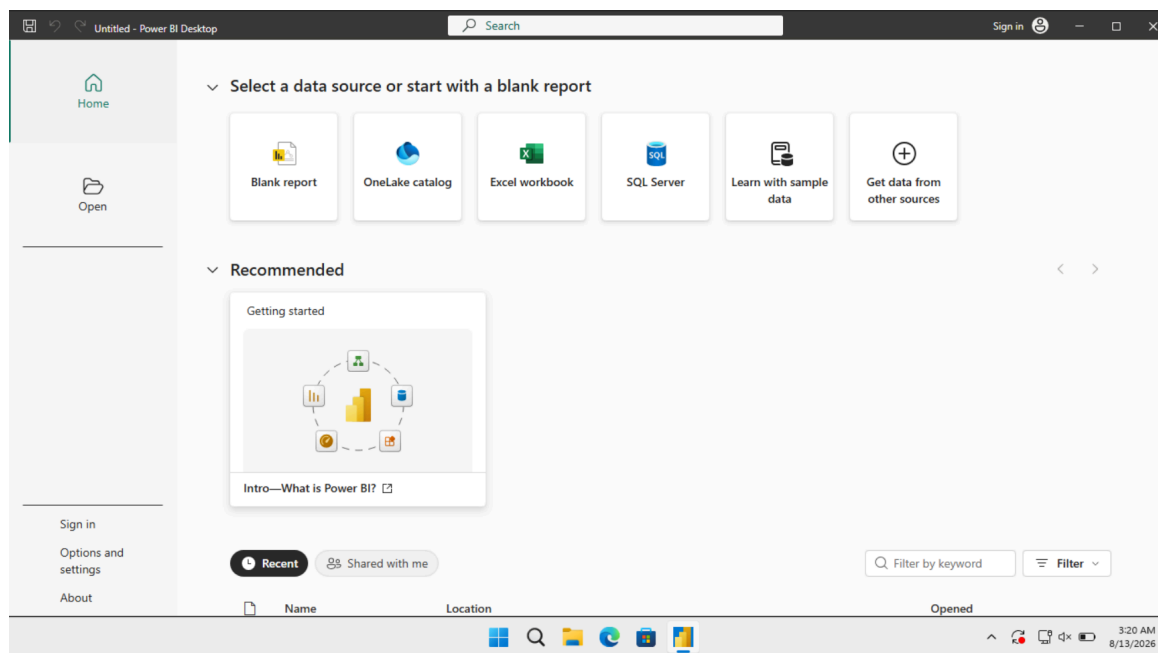


Step 2 – Open Microsoft Power BI Desktop

Microsoft Power BI Desktop was opened.

From the Power BI Home screen, the available data-source options were displayed.

The **Excel workbook** option was selected because the downloaded dataset was in Excel format.



Step 3 – Select the Excel Dataset

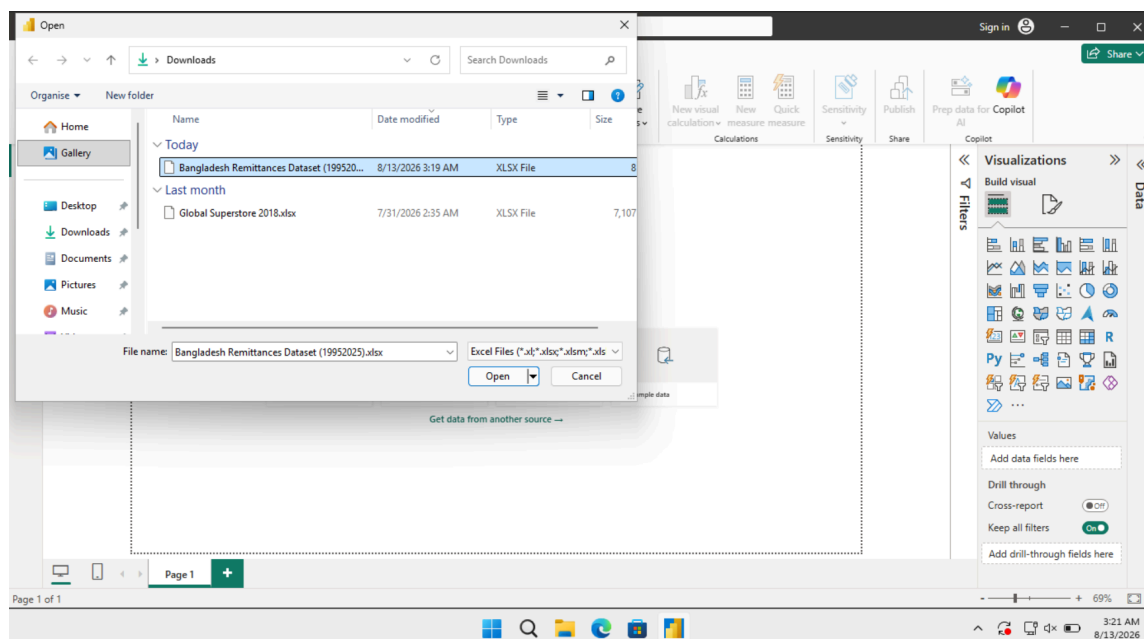
The downloaded Excel file was located in the **Downloads** folder.

The file:

Bangladesh Remittances Dataset (19952025).xlsx

was selected and opened in Power BI.

Power BI then analyzed the workbook and displayed the available sheet/table in the Navigator window.



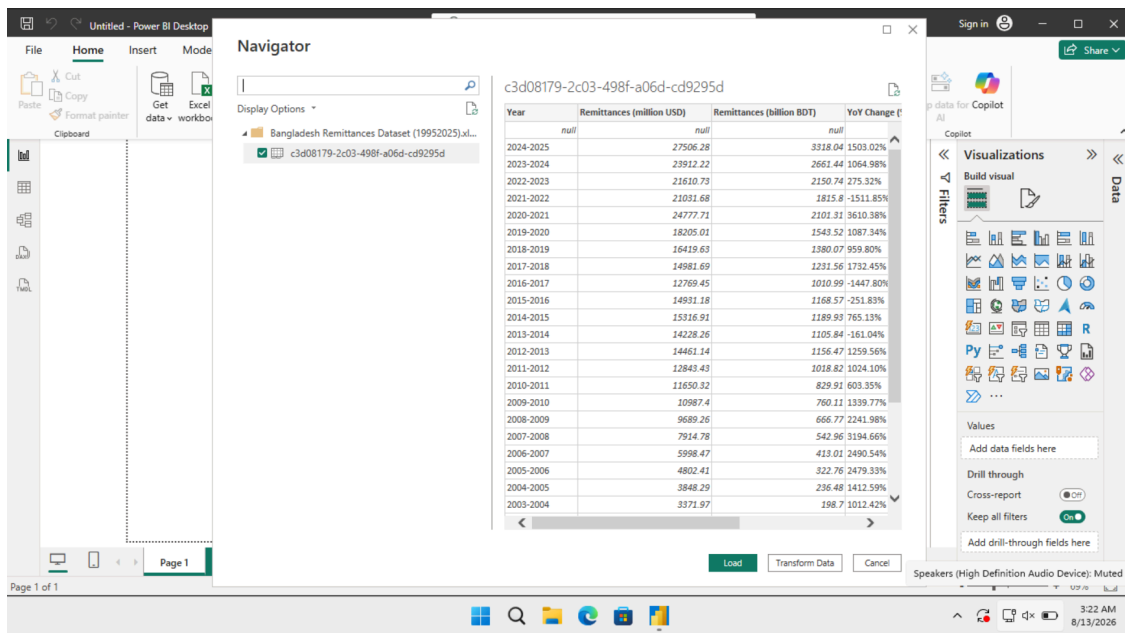
Step 4 – Preview and Import the Dataset

In the **Navigators** window, the available dataset table was selected.

The preview showed columns such as:

- Year
- Remittances (million USD)
- Remittances (billion BDT)
- YoY Change (%)
- Cumulative Growth vs. 1995–1996 (%)

Instead of directly loading the raw data, **Transform Data** was selected to clean and prepare the dataset using Power Query.



Step 5 – Clean and Transform the Main Dataset

The dataset was opened in **Power Query Editor**.

The following transformations were performed:

1. Promoted the first row as headers.
2. Checked the data types of the columns.
3. Removed duplicate records.
4. Removed blank rows.
5. Verified that the columns contained valid values.
6. Confirmed that the dataset contained **30 unique yearly records**.
7. Checked the numerical columns for errors.

The cleaned main table was named:

BRC-1995-2025

The table contained five important columns:

- Year
- Remittances (million USD)
- Remittances (billion BDT)
- YoY Change (%)

- Cumulative Growth vs. 1995–1996 (%)

Table: SelectRows(#"Removed Duplicates", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"", " "}))

Year	Remittances (million USD)	Remittances (billion BDT)	YoY Change (%)	Cumulative Growth vs. 1995-1996 (%)
2024-2025	27506.28	3318.04	1503.02%	216005.95%
2023-2024	23912.22	2661.44	1064.98%	186475.28%
2022-2023	21610.73	2150.74	275.32%	167565.03%
2021-2022	21031.68	1815.8	-1511.85%	162807.26%
2020-2021	24777.71	2101.31	3610.38%	193586.59%
2019-2020	18205.01	1543.52	1087.34%	139581.86%
2018-2019	16419.63	1380.07	959.80%	124912.25%
2017-2018	14981.69	1231.56	1732.45%	113097.38%
2016-2017	12769.45	1010.99	-1447.80%	94920.46%
2015-2016	14931.18	1169.57	-251.83%	112682.37%
2014-2015	15316.91	1189.93	765.13%	115851.72%
2013-2014	14228.26	1105.84	-161.04%	106906.81%
2012-2013	14461.14	1156.47	1259.56%	108820.27%
2011-2012	12843.43	1018.82	1024.10%	95528.32%
2010-2011	11650.32	829.91	603.35%	85725.11%
2009-2010	10987.4	760.11	1339.77%	80278.21%
2008-2009	6220.65	666.77	3221.00%	69612.02%

Step 6 – Create the Remittance Table

A separate query/table named **Remittance** was created.

Only the relevant columns were retained:

- Year
- Remittances (billion BDT)
- Cumulative Growth vs. 1995–1996 (%)

This table can be used for analyzing remittance values in BDT and cumulative growth.

Table: SelectColumns(Source,{"Year", "Remittances (billion BDT)", "Cumulative Growth vs. 1995-1996 (%)"})

Year	Remittances (billion BDT)	Cumulative Growth vs. 1995-1996 (%)
2024-2025	3318.04	216005.95%
2023-2024	2661.44	186475.28%
2022-2023	2150.74	167565.03%
2021-2022	1815.8	162807.26%
2020-2021	2101.31	193586.59%
2019-2020	1543.52	139581.86%
2018-2019	1380.07	124912.25%
2017-2018	1231.56	113097.38%
2016-2017	1010.99	94920.46%
2015-2016	1169.57	112682.37%
2014-2015	1189.93	115851.72%
2013-2014	1105.84	106906.81%
2012-2013	1156.47	108820.27%
2011-2012	1018.82	95528.32%
2010-2011	829.91	85725.11%
2009-2010	760.11	80278.21%
2008-2009	666.77	69612.02%

Step 7 – Create the Amount Table

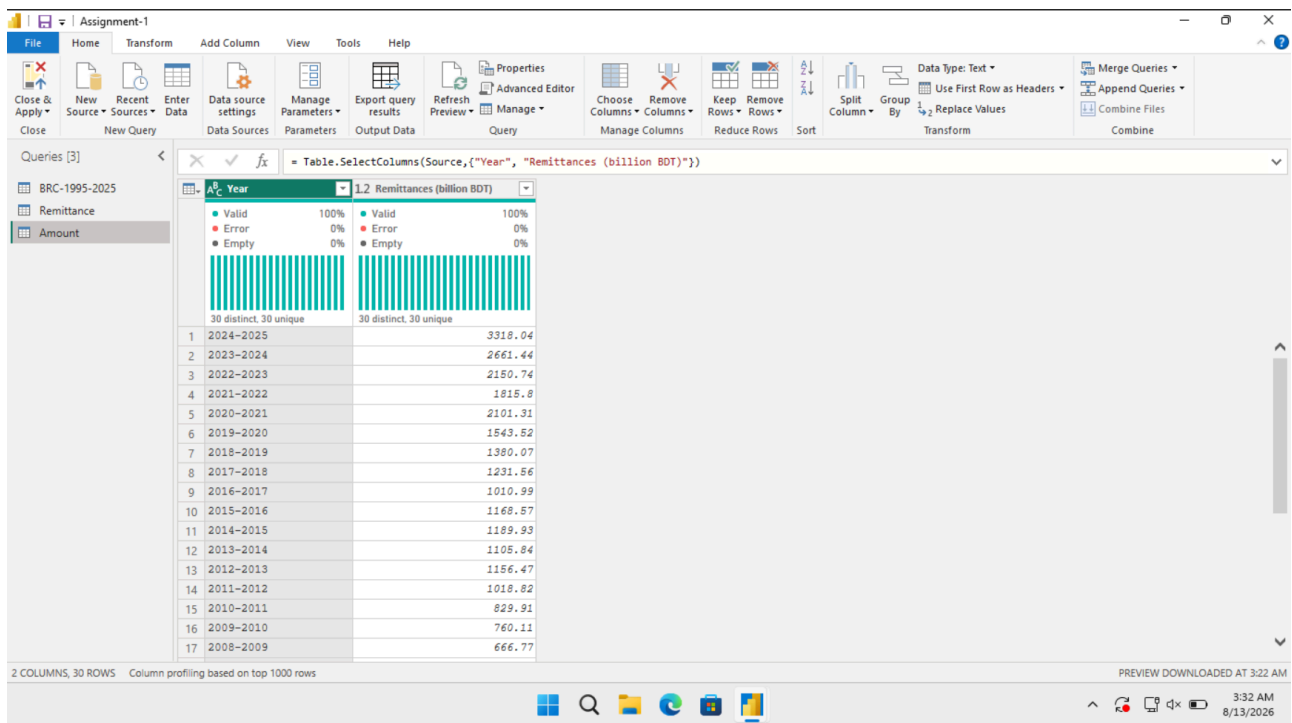
Another table named **Amount** was created.

The required columns were retained:

- Year
- Remittances (billion BDT)

This table provides a simplified structure for analyzing the yearly remittance amount in billion BDT.

After completing the transformations, the data was loaded into the Power BI data model.



Step 8 – Create Relationships Between Tables

The **Manage Relationships** option in Power BI was used to establish relationships between the tables.

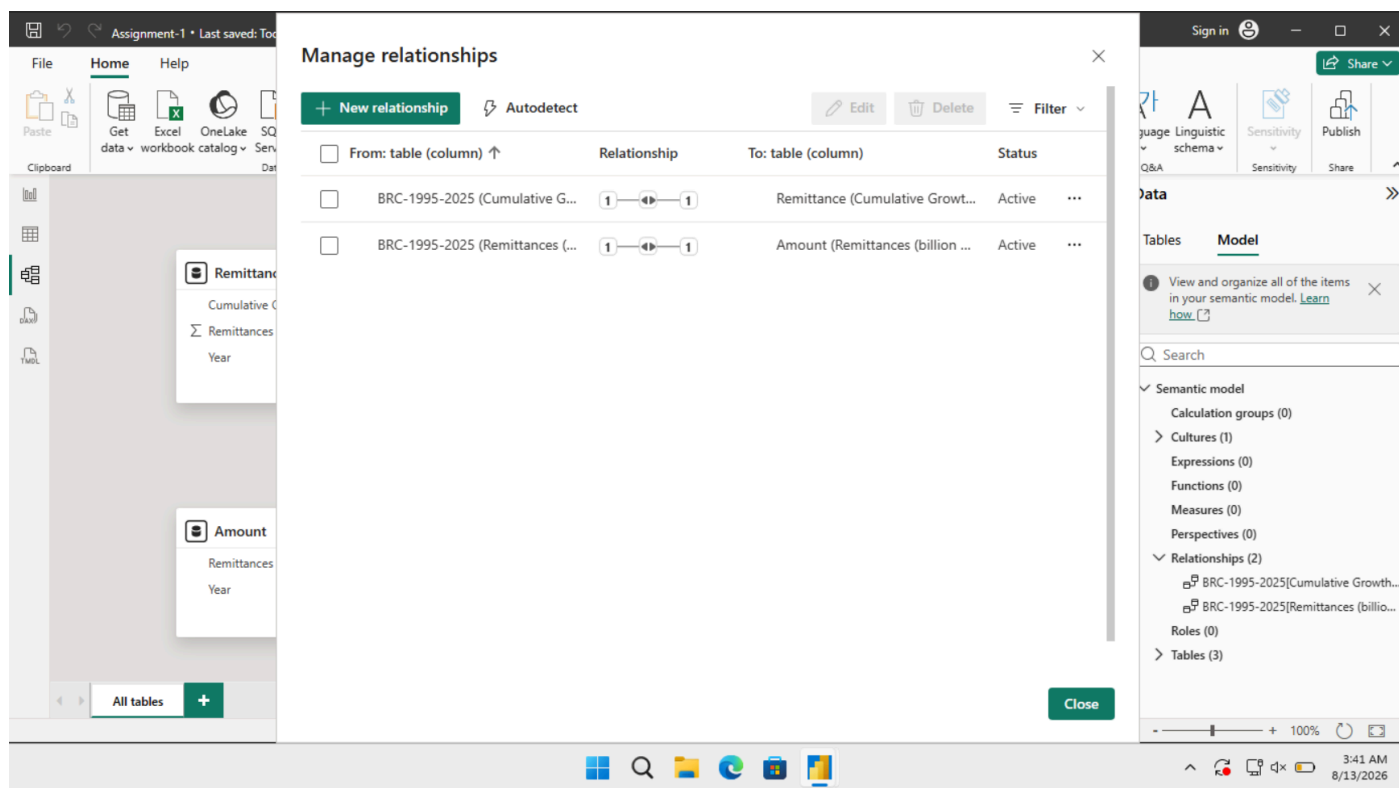
The common field used for connecting the tables was:

Year

Relationships were created between:

- **BRC-1995-2025** and **Remittance**
- **BRC-1995-2025** and **Amount**

The relationships were checked to ensure that the data model was connected correctly.



Step 9 – Check the Data Model

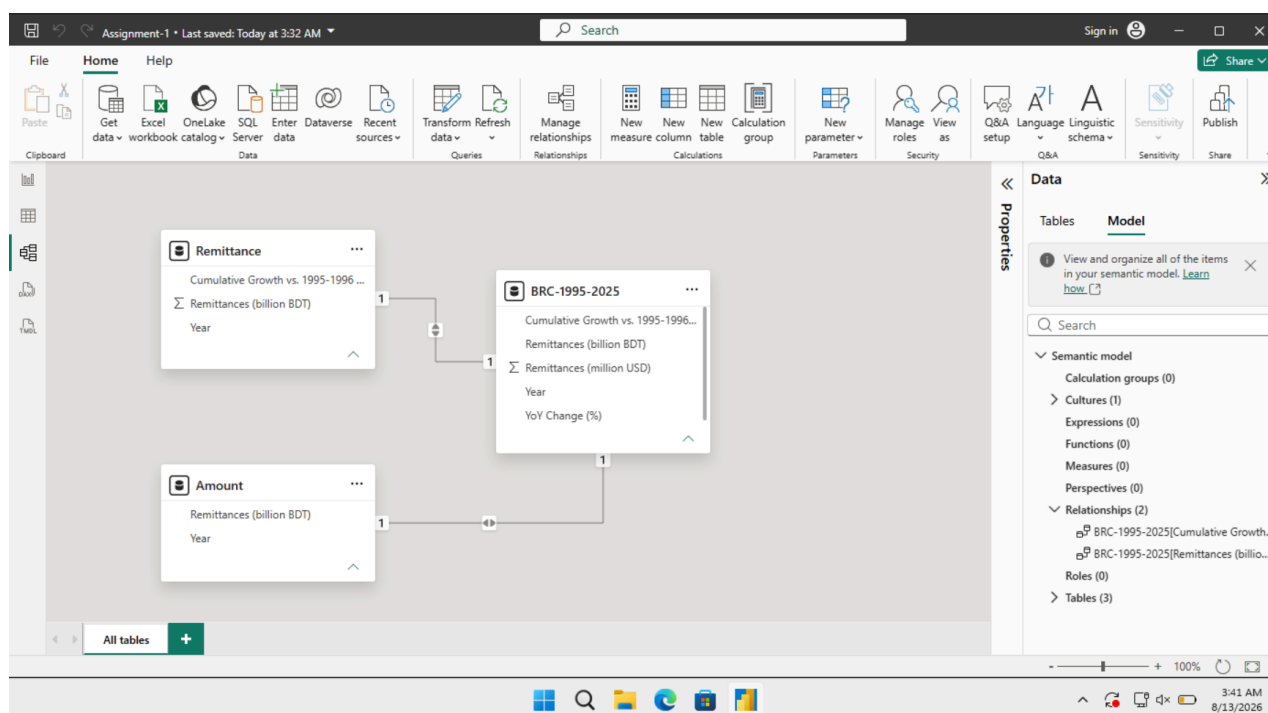
The **Model View** was opened to verify the structure of the data model.

The three tables were visible:

- BRC-1995-2025
- Remittance
- Amount

The relationship lines between the tables were verified.

This ensured that information from the different tables could be used together while creating Power BI visuals.



Step 10 – Create the Power BI Dashboard

After completing data preparation and modeling, the **Report View** was used to create the dashboard.

The following visualizations were created:

1. Card

A card was used to display the total remittances:

326.58K million USD

2. Funnel Chart

A funnel visualization was created to show the **Sum of Remittances (million USD) by Year**.

This provides a visual representation of the remittance values across different fiscal years.

3. Column Chart

A column chart was created to analyze:

Remittances (million USD) by YoY Change (%)

This helps identify variations in yearly growth.

4. Pie Chart

A pie chart was created to visualize:

Remittances (million USD) by Remittances (billion BDT)

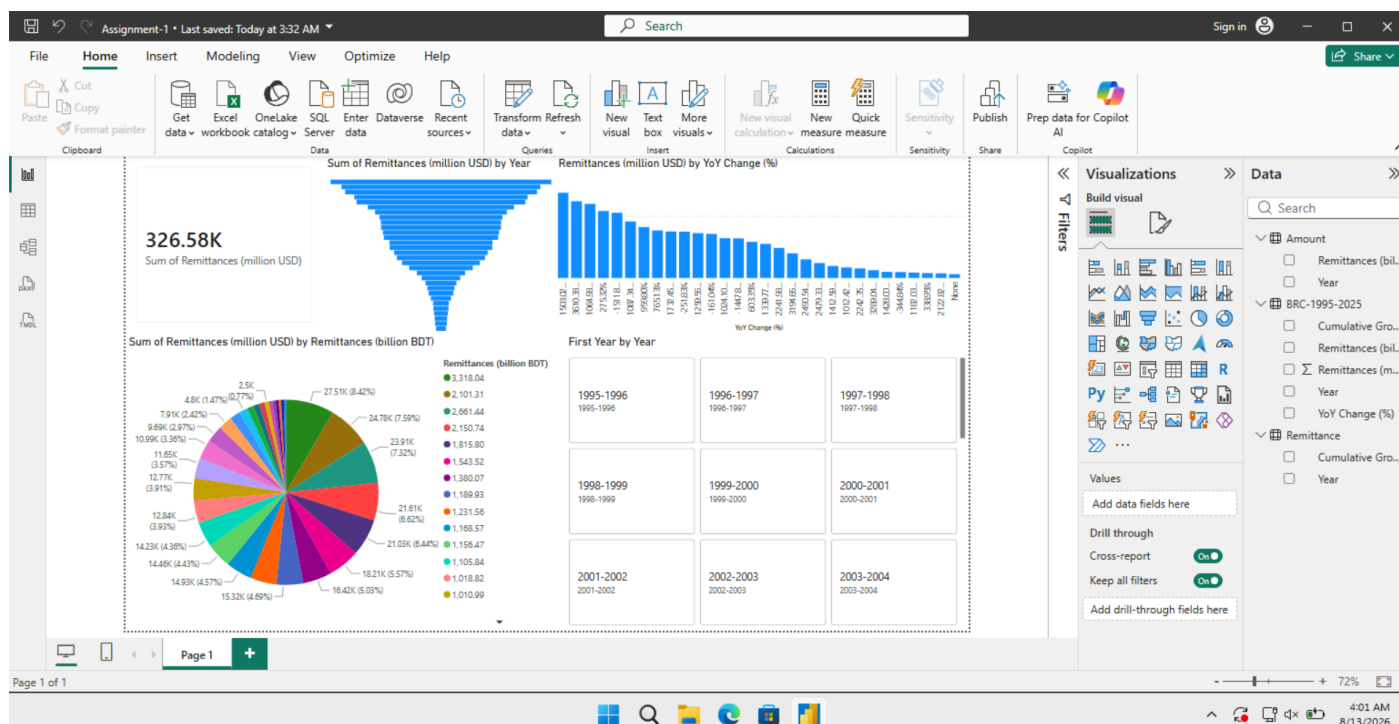
It provides a comparative view of yearly remittance contributions.

5. Year-wise Interactive Visual

A year-based visual was added to allow users to explore individual fiscal years such as:

- 1995–1996
- 1996–1997
- 1997–1998
- 1998–1999
- ...
- 2024–2025

The final dashboard combines these visuals into a single report page for interactive analysis.



Result

The **Bangladesh Remittances Dataset (1995–2025)** was successfully imported into Microsoft Power BI, cleaned using Power Query, divided into analytical tables, and connected through relationships using the **Year** field.

An interactive Power BI dashboard was successfully created containing:

- Total remittance KPI
- Year-wise remittance analysis
- YoY growth analysis
- Remittance comparison using BDT
- Interactive year-wise analysis
- Data model with connected tables

The dashboard provides a clear visual representation of the changing remittance inflows to Bangladesh over approximately three decades.

Learning Outcomes

After completing this practical, I was able to:

1. Understand the basic workflow of Business Intelligence.
2. Import an Excel dataset into Power BI.
3. Use Power Query for data cleaning and transformation.
4. Remove duplicate and blank records.
5. Create separate analytical tables from a dataset.
6. Understand data types and data quality validation.
7. Create relationships between Power BI tables.
8. Understand the Power BI data model.
9. Create KPI cards and different types of charts.
10. Build an interactive dashboard from real-world data.
11. Analyze yearly remittance trends and growth patterns.
12. Understand how raw data can be transformed into meaningful business insights.

Conclusion

This practical demonstrated the complete **Business Intelligence process using Microsoft Power BI**, starting from obtaining a real-world dataset and ending with an interactive analytical dashboard. The Bangladesh Remittances Dataset was successfully cleaned, transformed, modeled, and visualized. The practical helped in understanding how Power BI can convert large amounts of raw data into meaningful visual information that supports analysis and decision-making.